

Standard Operating Procedure Bench Model Hand Brake

SOP: Tennsmith HBU48-16 (Bench Model Hand Brake)

The Tennsmith HBU48-16 is a 48-inch, 16-gauge capacity universal (box and pan) hand brake. It is designed for precise bending of mild steel, aluminum, and stainless steel using a series of removable fingers to accommodate various box or chassis dimensions. This is a high-leverage manual tool; its primary hazards involve extreme pinch points and sharp material edges.

Potential Hazards

- **Mechanical (Clamping):** The clamping bar exerts immense pressure; fingers caught under the clamping fingers can be crushed or severed.
- **Mechanical (Apron):** The heavy apron and counterweights (if equipped) present a crush hazard during the upward and downward swing.
- **Operational:** Sharp sheet metal edges, "burred" corners, and potential for ergonomic strain during heavy-gauge bending.
- **Structural:** If the bench mount is not properly secured, the machine can tip or shift during high-leverage operations.

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	Safety Goggles (ANSI Z87.1): Mandatory. A full face shield is highly recommended when opening the door at temperatures above 400°C.
Thermal Protection	Heavy-Duty Heat-Resistant Gloves: Must be rated for temperatures up to 1125°C. Standard welding gloves are insufficient for direct contact with the chamber.
Body Protection	Flame-resistant lab coat and long pants are mandatory to protect against radiant heat and accidental contact with hot surfaces.
Respiratory	N95 Mask: Recommended during cleaning or maintenance. The ceramic fiber insulation can convert to crystalline silica (cristobalite) over 1000°C, which is a known respiratory hazard.



Standard Operating Procedure Bench Model Hand Brake

Before Use

- **Capacity Verification:** Ensure the material thickness does not exceed the machine's rated capacity:
  - **Mild Steel:** 16 Gauge (1.5 mm)
  - **Stainless Steel:** 20 Gauge (1.0 mm)

- **Finger Alignment:** Verify all fingers are securely tightened to the clamping bar. If performing box bending, ensure the gap between fingers is sufficient for the return flanges.
- **Workspace Check:** Ensure the "swing zone" of the apron is clear of tools, materials, and other personnel.
- **Setback Adjustment:** Adjust the clamping bar (setback) for the thickness of the metal to ensure a clean radius and prevent damage to the fingers.

During Use

- **Hand Placement:** Keep hands on the designated handles at all times. **Never** place hands near the clamping fingers once the workpiece is positioned.
- **Clamping Pressure:** Use the dual clamping handles to secure the metal. Ensure the workpiece is flat and square against the back gauge or layout lines.
- **Bending Motion:** Pull the apron handles upward in a smooth, controlled motion. Do not "jerk" the handles, as this can lead to over-bending or mechanical failure.
- **Multi-Operator Safety:** This is a one-person machine. If a second person is helping support a large sheet, they must be clear of the apron before the operator begins the bend.
- **Counterweight Awareness:** Be mindful of the momentum of the apron. Do not release the handles until the apron has returned to its lowest (rest) position.

After Use

- **Lower the Apron:** Ensure the apron is fully lowered and the clamping handles are in the "open" position to prevent tension on the springs.
- **Cleanup:** Use a brush to clear metal slivers and scale from the bed and fingers.
- **Material Management:** Deburr all bent parts immediately. Secure excess sheet stock in the designated vertical or horizontal storage racks.



- **Housekeeping:** Sweep the floor around the bench to prevent slips on metal fines or scale.

Emergency Equipment and Procedures

- **Crush Injuries:** In the event of a crush injury, immediately release the clamping handles to free the limb.
- **Reporting:** Report all incidents, including "near-misses," bent fingers, or structural instability of the bench, to the Laboratory Project Supervisor immediately.